SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.7: Microorganisms — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 1.7: Microorganisms
Driving Question	DRIVING QUESTION: Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Noticing and Wondering: What Is Growing on Our Ugali and Chapati?**	Students examined photographs and (where available) physical samples of mould-covered ugali and chapati. They recorded observations using their senses (sight, smell — no touching) and noted colour, texture, and location of fuzzy patches. Teachers should note that learners will likely conflate all microorganisms as 'germs'; this lesson surfaces that misconception productively.	Learners distinguished between living and non-living causes of food change (e.g., drying out vs. mould growth) and identified that the fuzzy patches are caused by a living organism too small to see with the naked eye. They activated prior knowledge about 'bacteria' and 'viruses' and began to appreciate that multiple categories of microorganism exist. Key vocabulary introduced: microorganism, fungi, observation, phenomenon.	The fuzzy patches on the ugali and chapati are colonies of fungi (mould) — living microorganisms that grow on organic matter rich in starch and moisture. Unlike drying or burning, mould growth is biological: the organism feeds on the food, reproduces, and spreads. This distinguishes a biological change from a physical/chemical one. Teachers should use this contrast explicitly: 'Ugali left in the sun gets hard — that is a physical change. Ugali left covered overnight grows fuzz — that is a living organism at work.' This lesson anchors all future lessons to the driving question phenomenon.
Lesson 2 **The Microbial World: Classification and Characteristics of Microorganisms**	Students observed labelled diagrams, micrographs, and short video clips (or textbook illustrations) showing bacteria, fungi, viruses, and protozoa at different magnifications. They completed a comparison chart noting cell type (prokaryote/eukaryote/acellular), size, and examples relevant to Kenya (e.g., Plasmodium — malaria; Mycobacterium tuberculosis; Saccharomyces — used in	Learners classified microorganisms into four major groups — bacteria, fungi, viruses, and protozoa — and identified the defining structural characteristics of each. They recognised that the mould on the ugali and chapati belongs to the fungi group, connecting back to Lesson 1. Key vocabulary: prokaryote, eukaryote, acellular, pathogen, saprophyte.	Microorganisms are classified by cell structure and biological organisation. Bacteria are unicellular prokaryotes (no membrane-bound nucleus); fungi are eukaryotes that may be unicellular (e.g., yeast) or multicellular (e.g., bread mould, Rhizopus stolonifer); viruses are acellular and can only replicate inside a host cell; protozoa are unicellular eukaryotes. The mould on the ugali is a multicellular fungus.

	mandazi and bread making). Teachers should cold-call at least 3 learners to describe one organism and justify its category.		Teachers should reinforce: 'Not all microorganisms are harmful — the same fungal kingdom that spoils our ugali also gives us the yeast that makes mandazi rise.' This dual nature foreshadows Lesson 6.
Lesson 3 **Modes of Transmission: How Did the Mould Get onto the Ugali?**	Students observed a teacher-led demonstration or simulation: a piece of bread/ugali left uncovered near an open window versus one sealed in a container — results shared after 24–48 hours (set up in a prior lesson). Learners also examined diagrams of airborne spore dispersal and reviewed results of a spore-settling experiment (agar plates or bread slices left open vs. covered in different locations in the school). Teachers should prompt: 'Which sample grew more mould? What was different about the conditions?'	Learners identified airborne transmission as the primary route by which fungal spores reached the uncovered ugali and chapati. They generalised that different microorganisms use different transmission routes: airborne/droplet (TB, COVID-19, mould spores), contact (ringworm, staphylococcal skin infections), vector-borne (malaria via Anopheles mosquito), waterborne (cholera, typhoid), and foodborne (Salmonella). Key vocabulary: transmission, spore, vector, airborne, droplet, contamination.	Mould spores are microscopic reproductive units released into the air by mature fungal colonies. When they land on a suitable substrate — warm, moist, starchy ugali or chapati left uncovered — they germinate and begin a new colony. This airborne route of transmission explains why covered food resists mould longer. Teachers should draw explicit comparisons: 'Just as TB bacilli travel in droplets when someone coughs on a matatu, mould spores travel in air currents in our kitchens.' This mechanistic reasoning directly answers the driving question and sets up prevention strategies in Lesson 5.
Lesson 4 **Infections Caused by Microorganisms: Fungi, Bacteria, and Viruses**	Students examined case studies of three patients presenting at a clinic in Kisumu: (1) a child with ringworm on the scalp (fungal), (2) a family member with cholera symptoms after drinking unboiled water from a river near Kisumu (bacterial), and (3) a student with influenza (viral). Learners identified the causative microorganism, category, transmission route, and symptoms for each case. Teachers should allow 3 minutes of pair discussion before cold-calling 4 pairs to share findings.	Learners catalogued human infections by microorganism type: fungal infections (ringworm/tinea, candidiasis, athlete's foot), bacterial infections (cholera — Vibrio cholerae, tuberculosis — Mycobacterium tuberculosis, typhoid — Salmonella typhi), and viral infections (influenza, HIV/AIDS, COVID-19). They recognised that the same fungal kingdom responsible for moulding their ugali also causes ringworm and other skin infections. Key vocabulary: infection, pathogen, symptom, causative agent, fungal, bacterial, viral.	Microorganisms cause disease when they colonise the human body and disrupt normal physiological function. Fungi (same kingdom as the ugali mould) cause superficial skin infections and, in immunocompromised individuals, systemic infections. Bacteria release toxins or trigger immune responses (e.g., Vibrio cholerae produces a toxin causing severe diarrhea). Viruses hijack host cells to replicate, destroying them in the process. Teachers should note: 'The fuzzy mould on your ugali and the ringworm on a child's scalp are caused by relatives in the same fungal kingdom — understanding one helps us understand the other.' This cross-context linkage is a critical pedagogical move.
Lesson 5 **Prevention and Control: How Mode of Transmission Shapes Our Defences**	Students matched prevention strategies to transmission routes using a Kenya-contextualised card-sort activity: covering food with a lid (airborne spores/droplets), boiling water (waterborne pathogens), using a mosquito net (vector-borne malaria), washing hands with soap (contact transmission), vaccination (viral infections).	Learners established the principle that the most effective prevention measure is matched to the mode of transmission of the specific microorganism. They identified key prevention strategies used in Kenyan households and public health contexts: food covering and refrigeration (mould/foodborne bacteria), water treatment and boiling	Because each mode of transmission represents a specific pathway for the pathogen to reach a new host, breaking that specific pathway is the most efficient control strategy. Covering ugali and chapati interrupts airborne spore deposition — directly preventing the phenomenon observed in the driving question. Boiling

	Teachers should circulate and probe: 'Why does a mosquito net prevent malaria but not cholera?' Allow 4 minutes, then whole-class debrief.	(waterborne), vector control and bed nets (malaria), hand hygiene (contact), vaccination (viral), and antibiotic/antifungal treatment (bacterial/fungal infections). Key vocabulary: prevention, control, vaccination, sterilisation, disinfection, antibiotic, antifungal.	water kills waterborne pathogens by denaturing their proteins. Mosquito nets create a physical barrier against the Anopheles vector. Teachers should make this logic explicit: 'Prevention is not random — it is targeted at the weakest link in the chain of transmission.' This mechanistic reasoning supports transfer to new disease contexts.
Lesson 6 **Economic Importance of Microorganisms: The Dual Nature of the Invisible World**	Students examined two sets of Kenyan economic scenarios: (1) Beneficial — a dairy cooperative in Eldoret using Lactobacillus to ferment mala (sour milk); a baker in Nairobi using Saccharomyces yeast to leaven bread and mandazi; a farmer in the Rift Valley using Rhizobium bacteria to fix nitrogen in bean fields; a hospital using Penicillium-derived antibiotics. (2) Harmful — post-harvest maize losses from Aspergillus mould (aflatoxin contamination) in coastal Kenya; cholera outbreaks affecting fishing communities near Lake Victoria; crop diseases. Teachers should ask: 'Which of these microorganisms is most closely related to the one on our ugali?'	Learners articulated the dual economic nature of microorganisms: beneficial roles include fermentation (food and beverage production), antibiotic manufacture, nitrogen fixation, and decomposition of organic waste; harmful roles include food spoilage, crop disease, and human/animal infections. They recognised fungi specifically as both the cause of ugali mould/aflatoxin contamination AND the source of penicillin antibiotics. Key vocabulary: fermentation, decomposition, nitrogen fixation, aflatoxin, antibiotic, economic importance.	Microorganisms are economically significant across agriculture, medicine, and food production in Kenya. The same fungal kingdom that spoils ugali (Rhizopus/Aspergillus) also produces life-saving antibiotics (Penicillium) and drives the fermentation that makes uji, mala, and mandazi possible. Aspergillus flavus produces aflatoxin — a serious food safety hazard in poorly stored Kenyan maize and groundnuts — illustrating that harmful effects extend beyond simple spoilage to toxic contamination. Teachers should emphasise: 'Our driving question asked why the ugali grew fuzzy patches — it also opens a window to understanding why Kenyan farmers lose millions of shillings of maize each year to the same fungal relatives.'
Lesson 7 **Laboratory Skills: Observing Microorganisms and Safe Handling Practices**	Students conducted a structured laboratory activity: preparing wet mount slides of yeast (Saccharomyces cerevisiae) suspended in sugar solution and — where available — examining prepared slides of bacteria (rod, coccus, spirillum forms) and fungal hyphae under light microscopes. Learners drew and labelled scientific diagrams at each magnification. Where microscopes are unavailable, teachers used high-resolution projected micrographs and had learners annotate printed diagrams. Safety briefing on handling biological material was conducted before the practical.	Learners applied correct microscopy technique (focusing from low to high magnification, preparing wet mounts, using cover slips) and practised biological drawing conventions (pencil, continuous lines, labelled with ruled lines). They identified bacterial cell shapes (cocci, bacilli, spirilla) and fungal structures (hyphae, spores) and connected microscopic structure to function. They also internalised biosafety rules for handling microbial specimens. Key vocabulary: hyphae, spore, cocci, bacilli, wet mount, magnification, biosafety.	Microscopy allows scientists to observe the microorganisms that are invisible to the naked eye but responsible for the phenomenon of mould growth on ugali. The structural features visible under the microscope — fungal hyphae forming the fuzzy mat, spores for reproduction — directly explain how mould spreads and grows. Teachers should connect back explicitly: 'The fuzzy patch you saw on Day 1 is a dense mat of hyphae like the ones you are now drawing.' This lesson reinforces the NGSS practice of using tools to observe and record data, and supports scientific communication through accurate biological drawing.
Lesson 8	Students revisited their original Lesson 1	Learners synthesised all unit learning into a	The driving question is now fully

Synthesis and Model Revision: Answering Our Driving Question	predictions and Driving Question Board entries. They reviewed all evidence gathered across the unit: mould observation (L1), classification (L2), spore transmission demonstration (L3), infection case studies (L4), prevention card sort (L5), economic analysis (L6), and microscopy drawings (L7). Working in groups, learners revised their cumulative explanatory model (diagram/poster) to produce a complete, evidence-based answer to the driving question. Teachers should cold-call 3 groups to present their model and facilitate peer critique using sentence starters: 'Your model explains… but does not yet account for…'	coherent, multi-causal explanation: the ugali and chapati grew fuzzy patches because airborne fungal spores (from the kingdom Fungi) landed on warm, moist, starchy food left uncovered, germinated, and formed visible hyphal colonies. They connected this phenomenon to human fungal infections, prevention strategies, and economic impacts on Kenyan food security. Learners also reflected on how their understanding changed from Lesson 1 to Lesson 8 — a metacognitive practice teachers should explicitly scaffold with a 'Then vs. Now' reflection prompt.	answerable with scientific evidence: fungi, specifically moulds such as Rhizopus stolonifer, are responsible for the fuzzy patches. Their airborne spores exploit uncovered, nutrient-rich food in warm Kenyan kitchen conditions. Understanding this invisible world empowers Kenyans to protect food through proper storage (covering, refrigeration, drying), to recognise fungal, bacterial, and viral infections and their appropriate treatments, and to appreciate the beneficial roles microorganisms play in food production and agriculture. Teachers should close by affirming: 'Science does not just explain the world — it gives us tools to protect our families, our food, and our livelihoods.'